

Differential Geometry Notes

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September 2026

1 Vector fields

Claim 1. *The map $X, Y \rightarrow D_X Y$ is bilinear and has the following properties:*

- $D_{fX} Y = f(D_X Y)$
- $D_X(fY) = f(D_X Y) + (D_X f)Y$
- $D_X Y - D_Y X = [X, Y]$
- $D_X \langle Y, Z \rangle = \langle D_X Y, Z \rangle + \langle Y, D_X Z \rangle$

Claim 2. *Suppose $\phi : \Omega \rightarrow V$ is a local parametrization. Then $D_{\phi_i} \phi_j = \frac{\partial \phi}{\partial x_i \partial x_j}$*

2 Riemannian metric

Definition 1. *Suppose $\gamma : [a, b] \rightarrow M$ is a piecewise smooth curve.*

A reparametrization of γ is given by $\gamma \circ \phi$, where $\phi : [c, d] \rightarrow [a, b]$ is a homeomorphism with a partition $c_0, c_1, \dots, c_k$ such that $\phi|_{[c_{i-1}, c_i]}$ is a diffeomorphism onto its image

Claim 3. *Suppose M is connected. Then there is a piecewise regular curve between any two points*

Definition 2. *Suppose M is connected. Then we define the Riemannian metric by*

$$d_M(p, q) = \inf\{L(\gamma) : \gamma \text{ is a piecewise smooth curve from } p \text{ to } q\}$$

Claim 4. *The topology induced by the Riemannian metric is the same as the subspace topology M inherits from $\mathbb{R}^k$*

Claim 5. *If there is a piecewise smooth curve from p to q of length $d_M(p, q)$ then there is such a piecewise regular curve as well. This is true for every p and q if M is closed in $\mathbb{R}^k$*

Definition 3. *A diffeomorphism $F : M \rightarrow N$ is called a Riemannian isometry if at each point the derivative DF_p is an isometry from $T_p M$ to $T_p N$*

Claim 6. Every Riemannian isometry is an isometry with respect to the Riemannian metric

Definition 4. A smooth map $F : M \rightarrow N$ is called a local Riemannian isometry at p if there is a neighborhood U such that $F(U)$ is open and $F|_U : U \rightarrow F(U)$ is a Riemannian isometry

3 First fundamental form

Definition 5. The first fundamental form of M at p is defined by restricting the standard inner product on $\mathbb{R}^k$ to the tangent space. We have $g_p : T_p M \times T_p M \rightarrow \mathbb{R}$

Claim 7. Two manifolds M and N are locally isometric around p and q if and only if there are local parametrizations $\phi : \Omega \rightarrow U$ and $\varphi : \Omega \rightarrow V$ such that $g_{\phi(a)}(v_i, v_j) = h_{\varphi(a)}(w_i, w_j)$ for all $a \in \Omega$

4 Covariant derivative

Definition 6. Suppose $X, Y \in \mathcal{X}(M)$ are smooth vector fields. Then we have the smooth vector field $\nabla_X Y = \pi \circ D_X Y$

Claim 8. Suppose $F : M \rightarrow N$ is an isometry then $\nabla_{F_* X}(F_* Y) = F_*(\nabla_X Y)$

Claim 9. The map $X, Y \rightarrow \nabla_X Y$ is bilinear and has the following properties:

- $\nabla_X Y - \nabla_Y X = [X, Y]$
- $\nabla_{fX} Y = f(\nabla_X Y)$
- $\nabla_X(fY) = f(\nabla_X Y) + (D_X f)Y$

5 Vector field along a curve

Definition 7. Suppose $\gamma : I \rightarrow M$ is a smooth curve. A smooth vector field along γ is a map $V : I \rightarrow \mathbb{R}^k$ such that $V(t) \in T_{\gamma(t)} M$

The vector space of smooth vector fields along γ is $\mathcal{X}(\gamma)$

Definition 8. We have $\text{cov}_\gamma \in L(\mathcal{X}(\gamma))$ given by $V \rightarrow (t \rightarrow \pi_{\gamma(t)} \dot{V}(t))$

6 Second fundamental form

Definition 9. The second fundamental form of M at p is the bilinear map $\Pi_p : T_p M \times T_p M \rightarrow \mathbb{R}^k$ given by $v, w \rightarrow [(D\pi)_p v]w$

Claim 10. $\text{range } \Pi_p \subseteq N_p M$ for all p

Claim 11. The second fundamental form is symmetric